

Sima Noorani

nooranis@seas.upenn.edu
[your-website-here]

Education

University of Pennsylvania Philadelphia, PA
Ph.D., Electrical and Systems Engineering
(Advisors: George J. Pappas, Hamed Hassani) 2022 – Present

University of Pennsylvania Philadelphia, PA
M.S.E., Electrical and Systems Engineering 2022 – 2025

Drexel University Philadelphia, PA
B.S., Electrical Engineering (Minor: Computer Science) 2017 – 2022

Work Experience

Cybersecurity Engineer Intern, Lockheed Martin, PA Apr. 2021 – Apr. 2022

Cloud Developer Intern, Bristol Myers Squibb, Hopewell, NJ Apr. 2020 – Sep. 2020

Machine Learning Engineer Intern, Comcast, Philadelphia, PA Apr. 2019 – Sep. 2019

Undergraduate Research Assistant, Drexel University Jun. 2018 – Sep. 2018

Resident Assistant, Housing and Residence Life, Drexel University 2018 – 2019

Multi-subject Tutor, Academic Center for Engineers, Drexel University Mar. 2018 – Apr. 2019

Research Interests

I am interested in developing uncertainty quantification methods for modern machine learning models—including generative models such as LLMs—and in designing guarantees that remain meaningful for end users making real decisions, especially in human–AI collaborative settings.

Keywords: Conformal prediction, uncertainty quantification, human–AI collaboration

Papers

Conference and Workshop Papers

- [1] **S. Noorani***, S. Kiyani*, G. J. Pappas, and H. Hassani. *Conformal Prediction Beyond the Seen: A Missing Mass Perspective for Uncertainty Quantification in Generative Models*. Neural Information Processing Systems (NeurIPS), 2025. Also accepted to the R2-FM Workshop at ICML 2025.
- [2] **S. Noorani**, O. Romero, N. Dal Fabbro, H. Hassani, and G. J. Pappas. *Conformal Risk Minimization with Variance Reduction*. In submission. Also accepted to the R2-FM Workshop at ICML 2025.

Preprints

- [1] **S. Noorani***, S. Kiyani*, G. J. Pappas, and H. Hassani. *Human-AI Collaborative Uncertainty Quantification*. arXiv:2510.23476, 2025.

Professional Activity

Reviewer

International Conference on Learning Representations (ICLR): 2025, 2026.

Learning for Dynamics and Control (L4DC): 2026.

IEEE International Symposium on Information Theory (ISIT): 2025.

Teaching Experience

Teaching Assistant

CIS 5200: *Foundations of Machine Learning* University of Pennsylvania

CIS 3333: *Mathematics of Machine Learning* University of Pennsylvania

Other Teaching & Mentoring

Multi-subject Tutor, Academic Center for Engineers Drexel University

Engineering Leadership Teaching Assistant/Scholar Drexel University

Instructor-Assistant Volunteer, TechGirlz Drexel University

Awards & Honors

Dean's Fellowship University of Pennsylvania, 2022 – Present

Dean's List Distinction Drexel University, 2017 – 2022

STAR Scholar Drexel University, 2017

Merit-based Founder's Scholarship Drexel University, 2017 – 2022

Honors Student Distinction Drexel University, 2017 – 2022

Selected Talks

- *Conformal Prediction Beyond the Seen: A Missing Mass Perspective for Uncertainty Quantification.*
Max Planck Institute, 2025.
- *Human-AI Collaborative Uncertainty Quantification.*
ANSR Reading Group, University of California, Berkeley, 2025.